Supplementary Information for "Percentile in Percentile: rating scholars against their career cohort by the impact of their papers"

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This supplement gives the detailed checks that the main text summarizes. Section, table, figure and equation numbers without an S refer to the main text.

S1 Factorization

Main text: Section 4.4.

PiP can be written as the author's productivity percentile times an impact multiplier. Let $F(m) = |\{b \in \mathscr{P} : n_b(0) \leq m\}|/|\mathscr{P}|$ be the productivity percentile. Unlike G_θ, it counts tied peers, so that it is positive for every member of the peer group. Defining the *impact multiplier* $Q_\tau(a) = \mathrm{PiP}_\tau(a)/F(n_a(0))$ gives

$$\mathrm{PiP}_\tau(a) = F(n_a(0)) \cdot Q_\tau(a). \tag{S1}$$

The identity holds by the definition of Q_τ; it describes a score rather than explaining it. The multiplier compares the author's standing on counts of articles above the cutoff with their standing on the number of articles, and it exceeds one when the author ranks higher among peers on the former. Because F counts ties and G_θ does not, an author whose standing is the same at every threshold has Q_τ slightly below one.

In our data $Q_{1/2}$ exceeds one for 30–36% of eligible authors, depending on career age. Across eligible authors with a positive score who started in 1975–2005, at career ages 5, 10 and 20, the variance of $\log \mathrm{PiP}_{1/2}$ splits into 0.74–0.82 from $\log Q$, about 0.30 from $\log F$ and a small negative covariance (-0.04 to -0.11). Most of the variation in PiP across authors thus comes from the impact multiplier rather than from the number of articles. These shares describe how scores vary, not how much of PiP reflects research quality.

S2 Ties

Main text: Section 4.4.

The standing counts peers with strictly fewer articles. Mid-ranks instead count half of the tied peers. They are a common choice for the percentile of a single paper (Bornmann et al., 2013; Waltman and Schreiber, 2013), but in the standing they would give credit to authors with nothing above a threshold: at high thresholds

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most peers have no article, and an author tied with them at zero would stand at half their share. In the simulated cohort, an author whose five articles all fall below the median would score 0.22 with mid-ranks instead of 0. The strict rule keeps weak articles from earning credit through ties; its price is that the mean score of a peer group depends on how common ties are (Eq. 6).

S3 The earlier, asymmetric form of the score

Main text: Section 4.5.

The earlier form of the score (Ipeirotis, 2024) is asymmetric; the website `https://www.pip-score.org` uses it. The asymmetric form, as the website computes it, sorts an author's articles by percentile, $y_{(1)} \geq y_{(2)} \geq \cdots$, and computes the area under the step curve that plots $y_{(k)}$ against the productivity percentile $F(k)$:

$$\text{PiP-AUC}(a) = \sum_{k=1}^{n_a(0)} y_{(k)} \left[F(k) - F(k-1)\right] = \int_0^1 F(n_a(\theta))\, d\theta. \tag{S2}$$

The second expression shows the difference from PiP: the asymmetric form compares the author's count at or above θ with peers' *total* counts rather than with their counts at or above θ.

The asymmetric form is the product of $F(n_a(0))$ and a weighted mean of the author's own percentiles, which keeps it bounded and monotone. But the weight of the k-th best article, $F(k) - F(k-1)$, is the share of peers with exactly k articles, which is zero for every k below the smallest article count among peers. Among the eligible authors of a cohort, nobody has fewer than three articles, so an author's best two articles receive no weight at all; among Google Scholar faculty profiles, the median author's best four articles receive none. This form is therefore dominated by productivity: the productivity percentile accounts for 0.59–0.77 of the variance of its logarithm. Scored against the simulated cohort, whose members all have at least three articles, an author with one article at the 99th percentile scores zero, while an author with 40 weak articles scores 0.10. Applying a cutoff τ to this form causes a second problem: it then compares partial counts with growing total counts, and its median at career age 10 halves between the 1960 and 2010 cohorts. The symmetric comparison in PiP removes both problems.

S4 The exact integral

Main text: Section 4.6.

To check that the classes do not drive the results, we also computed the exact integral for the 5.3 million eligible authors at career age 10 (start years 1960–2015). Write $y_{(k)}$ for the author's k-th best percentile. On $(y_{(k+1)}, y_{(k)}]$ the author has k articles at or above θ, and a peer has fewer exactly when its own k-th best percentile is below θ, so the integral reduces to sorting the peers' k-th best percentiles. Against the exact score, the 40-class score is 0.010 higher on average at $\tau = 0.5$ (0.055 at most) and ranks authors within cohorts almost identically: the rank correlation is 0.9994, and 96% of each cohort's top 10% is the same (start years 1975–2010). At $\tau = 0.75$ and 0.9 the correlation is 0.998 and 0.996 and the overlap 93% and

85%. None of the prediction results at career age 10 (Section 5.5) changes by more than 0.004 at any of the three cutoffs, so we keep the 40 classes.

S5 Retrospective check of the cutoff

Main text: Section 5.2.

The default was chosen from results that pooled all cohorts, so we checked afterwards whether the choice depends on which cohorts it is made from. Because the check was made after the default was chosen, it does not protect against specification search as an untouched holdout would. We split the prediction settings that compare authors of one career age into a choice sample (start years 1975–1990) and a test sample (start years 1991 and later). Stated as a rule, the default is the cutoff with the highest AUC for a future top-1% paper among cutoffs whose distribution is stable across cohorts ($\tau \leq 0.6$). Applied to the choice sample alone, the rule selects $\tau = 0.6$ at career age 10, with $\tau = 0.5$ within 0.001. The choice cohorts differ from the test cohorts in two respects. The knee of their quantity–impact frontier (Figure 6) lies at a slightly higher cutoff (between 0.6 and 0.75 rather than at 0.5). In those cohorts at career age 5, the top-10% count ties $\tau = 0.5$ on the top-1% outcome and beats it on future citation impact, whereas $\tau = 0.6$ keeps its lead.

In the test cohorts both cutoffs keep every comparison reported in Sections 5.5–5.6 at career ages 5, 10 and 15: they beat the h-index on all three outcomes, no other indicator beats either of them on more than one outcome, and both have the highest AUC for a future top-1% paper. Across τ from 0.25 to 0.6 that AUC varies by less than 0.01 in the test cohorts, so the exact choice in this range matters little for that outcome.

S6 Replication across cohorts

Main text: Section 5.5.

With more than a million authors per setting, sampling error is small: the standard error of a Spearman correlation of 0.4 would be below 0.001 for independent observations. Authors share papers, so the observations are not independent and this standard error is optimistic. The more relevant question is whether the differences hold in different populations. We repeated the age-10 comparison separately for each start year from 1975 to 2000, 26 cohorts of 17,867 to 122,114 authors. PiP beats the h-index on all three outcomes in all 26 cohorts, by 0.06–0.10 on the number of future top papers. It beats the top-10% count on the number of future top papers (by 0.05–0.08) and on the top-1% outcome in all 26 cohorts, and falls behind it on future citation impact in 20 (by up to 0.07). Against I3 and the article count, PiP wins on the top-1% outcome and on future citation impact in every cohort, and on the number of future top papers in 12 and 11 of the 26, with differences between -0.03 and $+0.04$.

S7 Early detection under the three-article rule

Main text: Section 5.6.

Requiring an h-index of three by career age 3 keeps only 21% of the authors who have three articles

by then, and it selects the population by one of the indicators being compared. We therefore repeated the analysis for all 1,455,891 authors with at least three articles at career age 3, with peers defined by the same rule. PiP still leads the *h*-index at every measurement age: the top 10% by PiP contains 24% future stars at age 3 against 20% for the *h*-index, and 31% against 27% at age 5. PiP reaches the *h*-index's age-10 accuracy 1.2 to 1.7 years earlier. It also leads the indicators of Table 6 on precision and on the AUC for a future top-1% paper at every age, but I3 matches or exceeds it on the AUC for future stars from about age 6.

S8 Agreement on the Google Scholar profiles

Main text: Section 5.7.

A last check measures agreement rather than prediction. On the Google Scholar profiles, where scores and honors are measured at the same time, the indicators are close: PiP is slightly ahead of the *h*-index for holding an honor (+0.004 [0.001, 0.008]) and for having a Wikipedia article (+0.012 [0.009, 0.015]). Total citations and the top-10% count do as well for holding an honor (their paired differences from PiP include zero) and slightly better for having a Wikipedia article (by 0.005 and 0.004).

S9 Author disambiguation

Main text: Section 5.9.

Semantic Scholar's author records come from automatic disambiguation, which can split one person across several records or merge different people. To check that the results are not artifacts of this algorithm, we use the author-curated Google Scholar profiles. We matched the 16,218 profiles to Semantic Scholar through their publications. A profile's work matches a Semantic Scholar paper on its normalized title (at least 25 characters) and a publication year within one year. The profile's counterpart is the Semantic Scholar author record with the same surname that holds the most matched works. We find a confident counterpart (at least five matched works and at least 30% of the profile's titles) for 14,511 profiles (89%). For the median person, 76% of their Google Scholar works are found in Semantic Scholar, and 90% of those found sit under the counterpart. Splitting is common, though: 64% of people have at least two of their works under a second record with the same surname and first initial, and 42% under a third. The counterpart is smaller than the profile (a median 67% of the profile's cited works, and 78% of its *h*-index), and its start year is later than the profile's for 47% of people, because early work is missing or sits under another record.

Despite these differences, the two sources rank people similarly. For the 9,796 people who have scores in both sources (Semantic Scholar start year 1995 or later, evaluated in 2025), the Spearman correlation between sources is 0.82 for the *h*-index and 0.78 for PiP. Each source computes percentiles against its own population: all eligible Semantic Scholar authors, and the Google Scholar profiles. For PiP, the agreement is lower for people split across three or more records (0.73) than for people with at most one record (0.76), but not by much. We read these comparisons as evidence that the results above are not artifacts of one disambiguation algorithm, and as a warning that individual Semantic Scholar scores can be off when a person's work is split.

S10 Survivorship

Main text: Section 5.9.

The population includes every eligible author, not only those who stayed in research, and the prediction results of Section 5.5 count authors who stop publishing as having no future top papers. This choice favors indicators that predict survival, that is, whether an author continues to publish. Measured at career age 5, the indicators predict whether an author is still publishing at career ages 11–15 with an AUC of 0.68 for the article count, 0.62 for PiP, 0.58 for the h-index and 0.54 for citations. Continuing to publish is a matter of volume, so the indicators that weigh volume most predict it best. PiP still leads the article count, the best predictor of survival, on the top-1% outcome and on future citation impact.

S11 The quantity–impact frontier

Main text: Section 5.5, Figure 6.

With $\tau = 0$, PiP predicts at least as well as I3 on all three outcomes in every setting (with $\tau = 0.25$, in all but one); with $\tau = 0.75$ it does at least as well as the top-10% count on all three outcomes in every setting; and from $\tau = 0.25$ to 0.75 it does better than the h-index on all three. Three indicators lie on or beyond the frontier. The mean percentile lies beyond it in the direction of pure citation impact, where it gives up any information about quantity. The article count lies just beyond it in two settings (at age 15 and in 2000), in the direction of pure quantity. The top-10% count per year of career differs from the plain count only when seniorities are mixed, and it lies on the curve next to $\tau = 0.75$.

S12 The field-normalized score

Main text: Section 5.8.

In this version, each paper's percentile is computed within its publication year and the author's main field, and peers are restricted to authors with the same main field and start year. A paper that does not carry the author's main field keeps its pooled percentile. The rule also gives a coauthored paper different percentiles for coauthors with different main fields, a choice that a rule based on the paper's own fields would avoid.

We have not separated the sources of the remaining spread. Two candidates are that papers outside the author's main field keep their pooled percentile, and that fields differ in how often peers tie, which also shapes the distribution of the score.

The two versions answer different questions: how a scholar's future work will rank among all articles of its year, and how it will rank within its field. With outcomes measured on pooled percentiles, the pooled score predicts best (0.427 / 0.772 / 0.321 against 0.414 / 0.770 / 0.298); with field-normalized outcomes, the field-normalized score predicts best (0.407 / 0.772 / 0.316 against 0.405 / 0.769 / 0.300). Both remain far ahead of the h-index (0.333 / 0.717 / 0.209 on field-normalized outcomes).

References

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